

Decker Supramolecular Architecture-Derived Near-Infrared Luminescent Materials for Bioimaging Application

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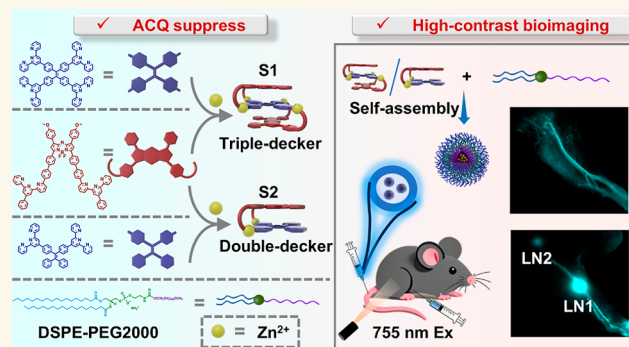
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Supporting Information

ABSTRACT: The in vivo performance of near-infrared (NIR) fluorescence imaging probes is governed by fluorophore aggregation behavior and molecular arrangement, with π – π stacking-induced aggregation-caused quenching (ACQ) being the key bottleneck limiting imaging efficacy. Conventional modification strategies are confined to monofunctional design, failing to synergistically achieve ACQ inhibition and spectral red shift. To address these issues, this study proposes an efficient synthetic strategy for decker structures, which relies on terpyridine steric hindrance design to enable precise, selective self-assembly and avoid multicomponent coassembly issues. We anchored aza-boron-dipyrromethene (aza-BODIPY) fluorophores and sterically hindered tetraphenylethylene (TPE) units onto terpyridine ligands. We designed and synthesized three systems with tailored TPE contents ($n_{\text{TPE}}/n_{(\text{BDP}+\text{TPE})}$): free aza-BODIPY BDP (0%), triple-decker S1 (33%), and double-decker S2 (50%). TPE-induced steric hindrance precisely modulates fluorophore arrangement and π – π stacking, enabling graded ACQ suppression while preserving the intrinsic NIR optical properties of aza-BODIPY. Among them, nanoparticles (NPs) formulated from S2 with the highest TPE content (50%) retain ~20% of the native brightness (9.3% for S1, while retention for BDP is negligible), achieving high-contrast deep-tissue NIR imaging (SBR = 5.6 at 1300 nm). This strategy provides an ideal model for elucidating fluorophore structure–property relationships and guides the design of high-performance NIR imaging probes.

KEYWORDS: supramolecular coordination complexes, ACQ suppression, NIR imaging, nanoprobe, vascular/lymphatic imaging



INTRODUCTION

NIR fluorescence imaging has revolutionized biomedical visualization by overcoming fundamental limitations of conventional optical techniques.^{1–7} Operating in the 700–1700 nm window, NIR light achieves deeper tissue penetration with superior resolution, thanks to reduced scattering/absorption and minimal autofluorescence that collectively enhance signal-to-noise ratios. The development of NIR-II imaging represents a quantum leap, where longer wavelengths enable unprecedented tissue penetration while virtually eliminating background interference.^{8–14} This technology also offers low phototoxicity, allowing prolonged imaging sessions, and is supported by ongoing advances in research methodologies and clinical applications—spanning diagnostics, image-guided interventions, and therapeutic monitoring.^{15–22}

Metallo-supramolecular complexes have emerged as a promising class of bioimaging agents, offering unique advantages over conventional molecular probes.^{23–30} Coordination-driven self-assembly provides an efficient approach for precisely organizing functional units with well-defined spatial arrangement and the stoichiometric ratio of different

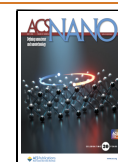
components, enabling systematic modulation of their intermolecular interactions and photophysical properties. The incorporation of classical imaging dyes further endows these supramolecular assemblies with optical traceability. For example, Mao's group constructed a metallacycle by integrating tetraphenylethylene-platinum (TPE-Pt) and perylenediimide (PPy) units for imaging-guided radiochemotherapy.³¹ The inherent structural flexibility allows for the incorporation of multiple functional components, including NIR fluorophores,^{32–38} targeting moieties,^{39–42} and responsive elements,⁴³ within a single discrete architecture. Despite these advantages, these systems still face the common challenge of insufficient spatial isolation between fluorophores, which limits

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Scheme 1. Schematic Illustration of Decker Supramolecular Architecture Design Strategies, Nanoparticle Preparation, and Applications in Bioimaging

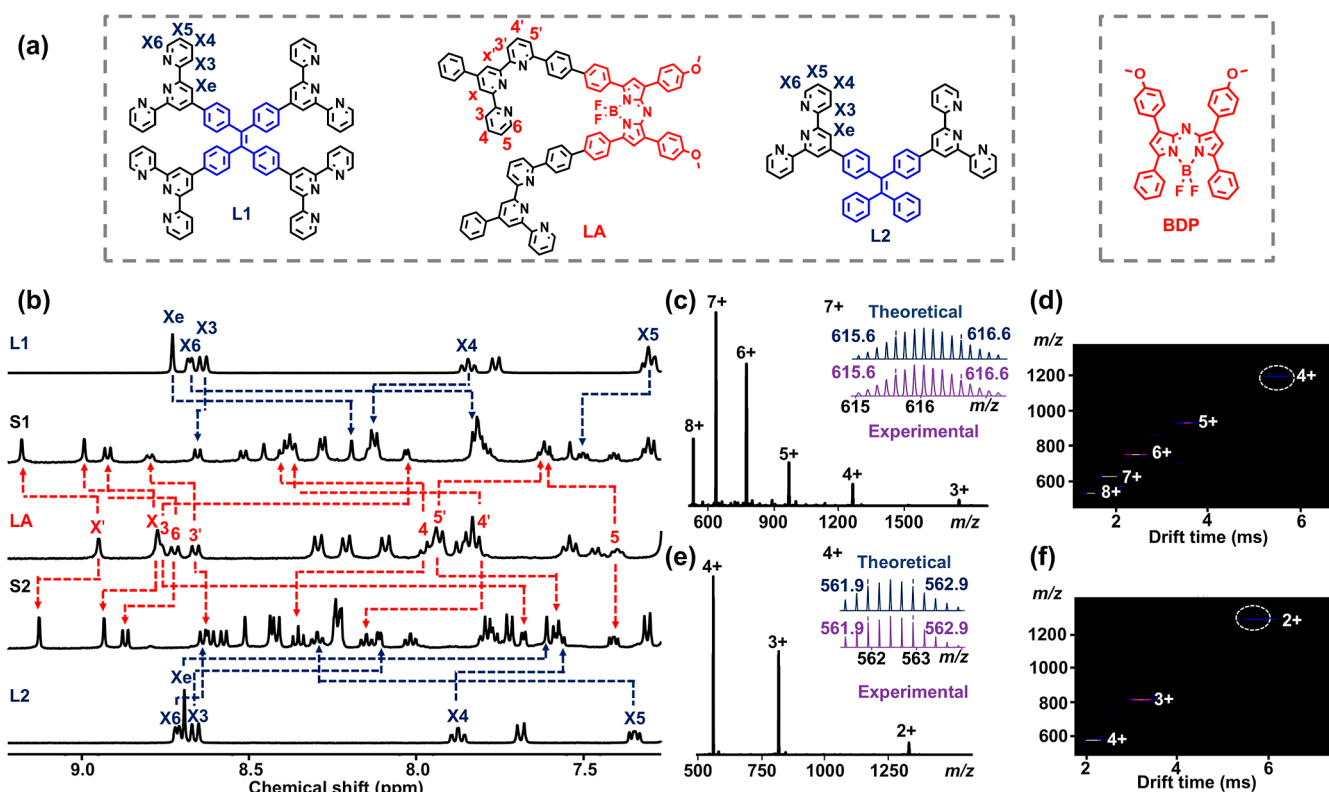
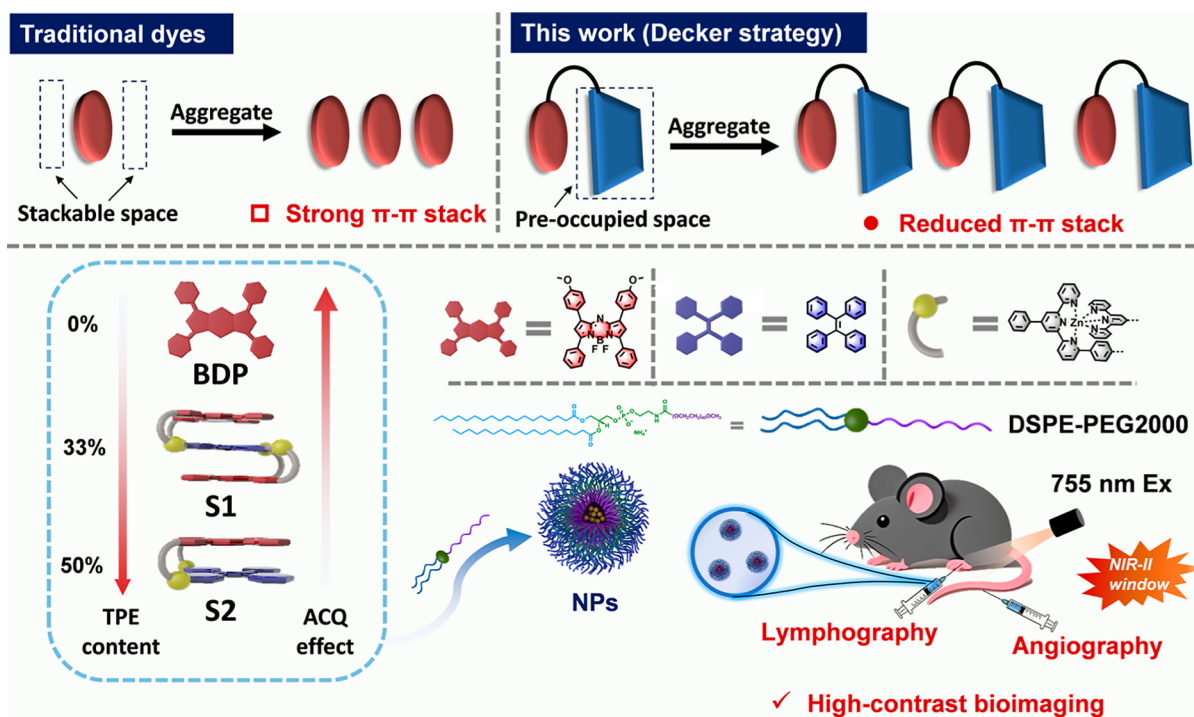


Figure 1. (a) Structure of ligands. (b) ^1H NMR spectra of ligands LA, L1, and L2 in CDCl_3 and complexes S1 and S2 in CD_3CN (500 MHz, 298 K). ESI-MS spectra of complexes (c) S1 and (e) S2. TWIM-MS spectra of complexes (d) S1 and (f) S2.

the efficacy of ACQ suppression in most supramolecular imaging probes.

To overcome this fundamental limitation, we have developed innovative decker supramolecular architectures,

triple-decker S1 (TPE content: 33%) and double-decker S2 (TPE content: 50%), which exhibit tailored steric hindrance effects relative to the unmodified BDP (TPE content: 0%). By introducing precisely controlled amounts of TPE moieties into

the terpyridine-based coordination framework, we enabled the ordered assembly of ACQ-sensitive aza-BODIPY fluorophores and sterically hindered TPE units (Schemes 1 and Figure 1a). This rational design strategy not only preserves the intrinsic NIR fluorescence of aza-BODIPY dyes but also induces tunable red-shifted absorption, with spectral red shifts of 50 nm for **S1** and 40 nm for **S2** relative to those of **BDP**. In vitro photophysical characterization further demonstrated that the distinct stacking modes between aza-BODIPY and TPE, dictated by the gradient TPE content, suppress the ACQ effect of aza-BODIPY to varying extents: **S2** (50% TPE) exhibits the most pronounced quenching inhibition, followed by **S1** (33% TPE), while **BDP** (0% TPE) shows a negligible ACQ resistance. Subsequent encapsulation of these supramolecular architectures with amphiphilic DSPE-PEG2000 yields biocompatible nanoparticles, which exhibit remarkably superior performance in real-time, noninvasive visualization of deep-tissue vasculature and lymphatic networks, thereby highlighting their advanced potential for preclinical imaging applications.

RESULTS AND DISCUSSION

The Synthesis and Characterization of Ligands and Decker Supramolecular Structures

The synthetic route of **LA** along with detailed characterization data is presented in Supporting Information (Figures S1–S5). In the design of ligand **LA**, the imaging moiety aza-BODIPY was incorporated into this ditopic ligand to facilitate subsequent further coordination for the synthesis of supramolecular structures. The 6-position modification of 2,2':6',2''-terpyridine (tpy) in **LA** is particularly noteworthy, as its characteristic claw-like geometry enables the construction of decker structures through shape-complementary self-assembly.^{44–46} The synthetic procedures and full characterization by ¹H and ¹³C nuclear magnetic resonance (NMR) spectroscopy, two-dimensional correlation spectroscopy (2D-COSY), and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry of all ligands are shown in Supporting Information (Figures S2–S5).

To achieve precise gradient control over the TPE content of the assemblies, we utilized TPE-based ligands with tailored tpy moiety counts. Specifically, ligand **LA** assembles with the tetratopic TPE ligand **L1** to form the triple-decker **S1**, while **LA** coordinates with the ditopic TPE ligand **L2** to yield the double-decker **S2**. For the synthesis of complexes **S1** and **S2**, the respective ligand combinations (**LA/L1** for **S1** and **LA/L2** for **S2**) were stoichiometrically mixed with Zn(NO₃)₂·6H₂O in CHCl₃/MeOH at 50 °C for 5 h to facilitate assembly. Upon addition of excess NH₄PF₆, a black precipitate formed and was directly used for subsequent characterization without further purification or isolation steps. NMR spectroscopy was employed to confirm the generation of the complexes. As illustrated in Figure 1b, both complexes showed sharp and well-resolved proton signals, indicating the formation of discrete assemblies. NMR resonances of complexes were further assigned by 2D COSY (Figures S6–S13). Electrospray ionization mass spectrometry (ESI-MS) revealed a series of prominent peaks with consecutive charge states, originating from the loss of anions (PF₆[−]). In addition, the experimental isotope patterns of each charge state closely matched the corresponding simulated patterns (Figures S14 and S15). These results provide compelling evidence for both molecular

weight and chemical composition of the structures, strongly supporting the formation of the expected heteroleptic complexes **S1** and **S2** (Figure 1c,e). To examine potential isomerism or conformational heterogeneity, traveling wave ion mobility-mass spectrometry (TWIM-MS) was conducted (Figure 1d,f) and narrow drift time distributions for all charge states were observed, which unambiguously demonstrated the formation of a single, well-defined product. Based on our group's previously reported decker assembly strategy, combined with NMR and ESI-MS characterization, the successful synthesis of the decker structures could be confirmed.

To further confirm the structures of the complexes, all attempts to grow single crystals of **S1** and **S2** were unsuccessful, and density functional theory (DFT) calculations were employed to optimize the ground-state configurations of **S1** and **S2** (Figure 2a,b). The results of the DFT analysis

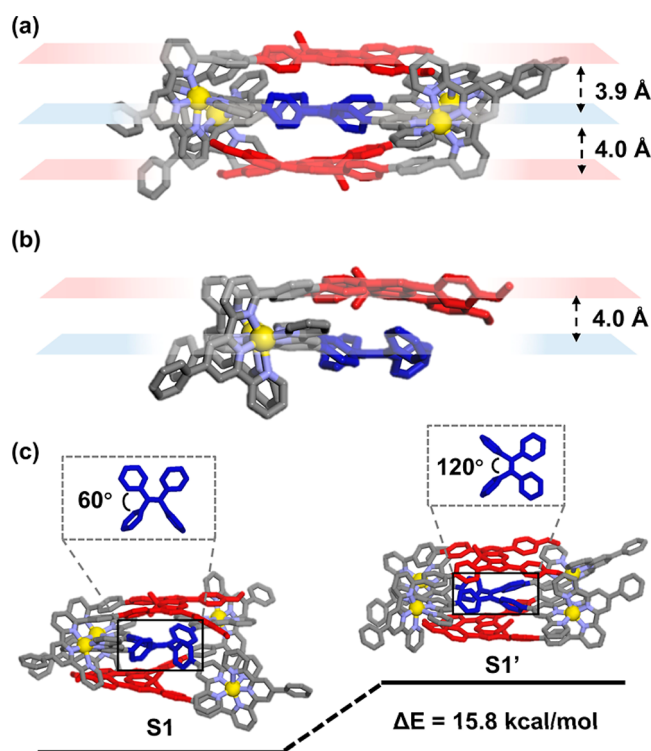


Figure 2. Optimized structures of complexes (a) **S1** and (b) **S2** by DFT calculation. (c) Calculated structures and energy difference of the two possible coordination modes of **S1** (hydrogen atoms are omitted for clarity).

indicated that triple-decker **S1** composed of one TPE unit and two aza-BODIPY moieties adopts a configuration wherein the central TPE spatially separates the two aza-BODIPY moieties. Similarly, double-decker **S2** consisting of one TPE unit and one aza-BODIPY moiety features one side of the aza-BODIPY unit sterically occupied by a TPE molecule. Benefiting from our decker supramolecular design strategy, each aza-BODIPY moiety incorporated into **S1** and **S2** is flanked by a TPE moiety on one side; this steric bulk occupies critical spatial domains and mitigates intermolecular stacking tendencies. In contrast, **BDP** lacks steric hindrance on both sides, rendering it far more prone to π – π stacking in the aggregated state. However, the polydentate binding capability and unique angular geometry of **L1** render two distinct coordination

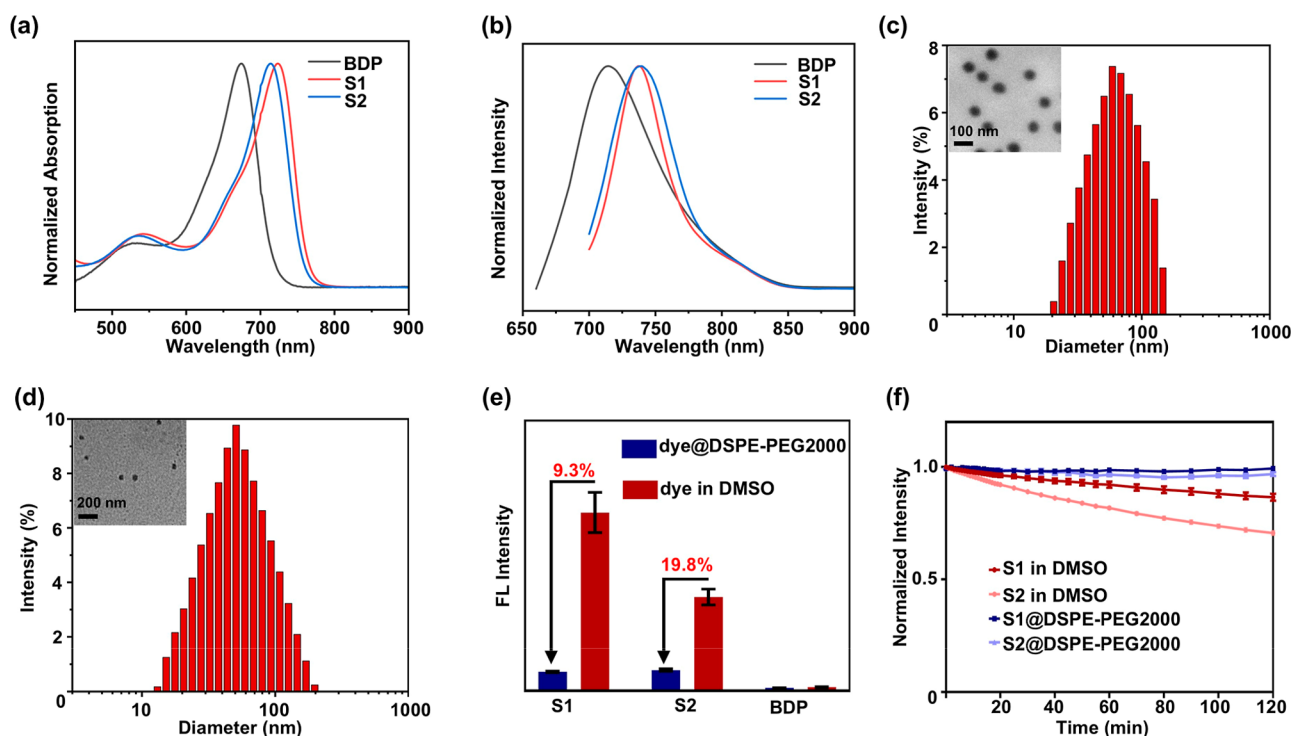


Figure 3. (a) Absorption and (b) fluorescent emission spectra ($\lambda_{\text{ex}} = 640$ nm) of BDP and complexes **S1** and **S2** at the concentration of $5 \mu\text{M}$ in DMSO. DLS result of (c) **S1@DSPE-PEG2000** and (d) **S2@DSPE-PEG2000** (inset: TEM images of NPs). (e) Brightness (mean $\pm$ SD, $n = 9$ independent samples per group, >900 nm, 6 ms) and (f) photostability (mean $\pm$ SD, $n = 3$ independent samples per group, >900 nm, 6 ms) of the dyes@**DSPE-PEG2000** and dyes in DMSO under a 755 nm excitation at a power density of $38 \text{ mW}/\text{cm}^2$.

modes feasible for the **LA** and **L1** assemblies: one involves the coordination of the two tpy moieties of **LA** with the two 120° -directed terpyridines of **L1**, while the other corresponds to the coordination of **LA**'s terpyridine units with the two 60° -directed terpyridines of **L1**. To unambiguously resolve the precise configuration of **S1**, DFT calculations were further conducted on both hypothetical structures (Figure 2c). The computational results indicated that the coordination mode between **LA**'s terpyridine moieties and **L1**'s 60° -directed terpyridines is thermodynamically favorable, as evidenced by the more optimal angular matching, more ordered molecular spatial stacking, and lower ground-state energy of the resulting complex, with an energy difference of 15.8 kcal/mol relative to the alternative mode.

Photophysical Properties of Decker Supramolecular Structures

The photophysical properties of the ligands and complexes were first investigated. Compared to the free aza-BODIPY molecule **BDP**, a characteristic absorption band emerged in the range of 580–740 nm with absorption peak at 673 nm (Figures 3a and S16). Upon incorporation with tpy moieties, this absorption band exhibited a significant red shift, with ligand **LA** displaying an absorption band in the range of 590–800 nm with peak at 720 nm (Figure S16), attributable to the extended π -conjugation. To probe the origin of this red-shifted absorption, the emission spectra of **LA** were recorded in solvents of different polarity. The emission maximum showed only minimal variation with solvent polarity (Figure S17), arguing against a dominant intramolecular charge transfer (ICT) character. Consequently, the red shift is primarily attributed to extended π -conjugation upon attachment of the tpy group. Similarly, the assembled structures **S1** and **S2** also

showed markedly red-shifted absorption peaks relative to **BDP** (**S1**: 723 nm; **S2**: 713 nm) (Figure 3a). Notably, the absorptions of the complexes extended into the near-infrared region, demonstrating superior NIR imaging potential compared to unmodified **BDP**. In DMSO, **S1** and **S2** also exhibited favorable emission properties, with maximum emission wavelength at 736 nm (**S1**) and 739 nm (**S2**) (Figure 3b), representing a significant red shift relative to **BDP**. Their photoluminescence quantum yields were determined to be 11.49% (**S1**) and 9.01% (**S2**), respectively (Figure S18a).

As a classic ACQ fluorophore, aza-BODIPY typically exhibits fluorescence quenching in the aggregated state, which limits its bioimaging applications. To systematically evaluate the anti-ACQ capability of the decker assemblies, fluorescence measurements were performed at three different concentrations (1, 5, and $10 \mu\text{M}$) in DMSO/ H_2O mixtures with varying water fractions (f_w), where DMSO acts as a good solvent and water as a poor solvent. For unmodified **BDP**, at all three concentrations, fluorescence intensity decreased progressively with increasing water content due to aggregation, consistent with the ACQ effect. When the water fraction reached 50%, the emission was nearly completely quenched (Figures S19 and S20). In contrast, both decker structures **S1** and **S2** maintained appreciable fluorescence at a 50% water content, demonstrating that our design effectively mitigates the ACQ effect. These results unequivocally demonstrate that the decker architecture effectively suppresses the ACQ effect over a broad concentration range. To further exclude the possibility that the observed anti-ACQ effect is specific to the DMSO/ H_2O system, we performed analogous fluorescence measurements in another mixed-solvent system, namely, acetonitrile/water ($\text{CH}_3\text{CN}/\text{H}_2\text{O}$), at a fixed concentration of $5 \mu\text{M}$ and

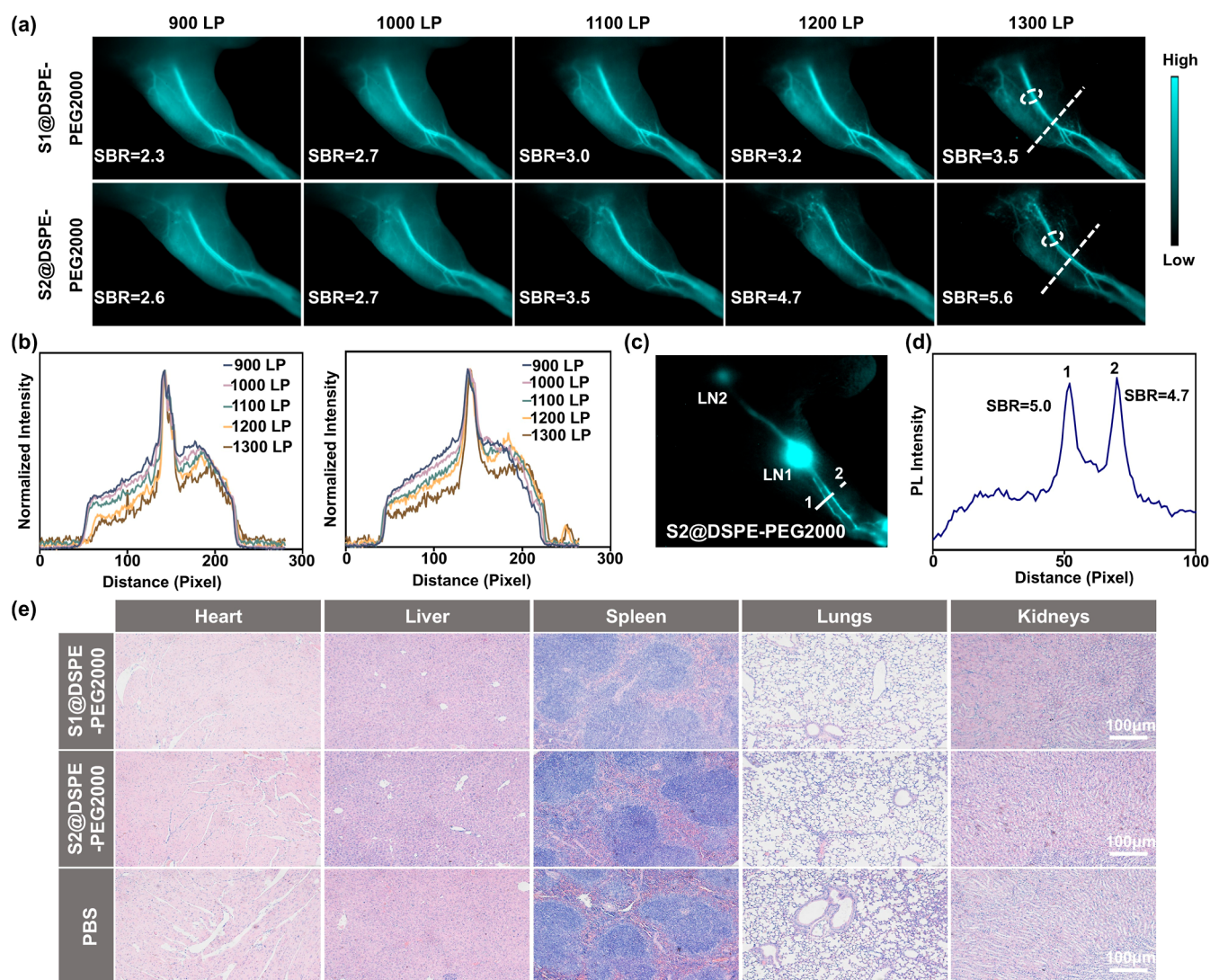


Figure 4. NIR-II angiography and lymphography of dyes@DSPE-PEG2000. (a) Hindlimb vessel imaging of dyes@DSPE-PEG2000 under NIR-II and tunable NIR-II windows (30/50/100/300/600 ms, 200 μ M, 200 μ L). (b) Cross-sectional fluorescence signals profile of the NIR-II hindlimb vessel imaging. (c) Lymphatic vascular imaging and lymph node imaging (>1200 nm, 550 ms, 200 μ M, 25 μ L) of S2@DSPE-PEG2000. (d) The corresponding cross-sectional intensity profile of the lymphatic vessels in the selected ROI and SNR statistics. LN1: popliteal lymph node. LN2: sacral lymph node. (e) H&E-stained images of tissue sections from different organs of mice treated with PBS (phosphate-buffered saline, control) or NPs. Scale bar: 100 μ m.

with varying water fractions. As shown in Figure S21, the control compound BDP again displayed a dramatic fluorescence drop upon increasing the water fraction to 50%, with almost complete quenching at $f_w = 50\%$. This is consistent with its severe ACQ behavior observed in DMSO/H₂O. In stark contrast, under the same high-water-content conditions ($f_w = 50\%$) in CH₃CN/H₂O, both S1 and S2 retained substantial fluorescence emission. These results demonstrate that the anti-ACQ capability of the decker architecture is not limited to DMSO/H₂O but represents a more general phenomenon. Structural analysis reveals that conventional aza-BODIPY molecules can undergo π - π stacking interactions through both planar faces. In our decker design, however, the propeller-shaped TPE moiety sterically occupies one side of the aza-BODIPY unit, thereby significantly suppressing intermolecular stacking and consequently enhancing the aggregated-state emission. To assess their performance in biologically relevant spectral regions, we quantified the brightness of S1 and S2 in DMSO across a series of distinct sub-NIR-II windows (Figure

S22). As a control, BDP was analyzed under identical experimental conditions, yielding only negligible fluorescence in DMSO. These results not only validate the enhanced optical properties of S1 and S2 relative to BDP but also underscore their greater applicability for in vivo imaging scenarios. Spectral overlaps were observed between the emission band of TPE-based L1 and L2 and the absorption band of aza-BODIPY-based LA, suggesting efficient Förster resonance energy transfer (FRET) within the decker complexes (Figures S18b,c). Although TPE exhibits AIE behavior,⁴⁷ its fluorescence is quenched via FRET. Consequently, neither emission peaks from L1 or L2, nor any AIE effect, were observed in the assembled system. To further probe the photophysical properties of the decker assemblies, the fluorescence lifetimes of S1 and S2 were measured in DMSO (Table S1 and Figure S23). Despite the structural difference in the TPE-based ligands (tetrapropyl for S1 vs ditopic for S2), the measured lifetimes are very close to each other. This result indicates that the emission of both

assemblies primarily originates from the aza-BODIPY core, which serves as the dominant emitter, while the TPE moieties play only a minor role in modulating the excited-state dynamics.

Preparation of Nanoprobes and High-Contrast NIR-II Imaging

The inherent hydrophobicity and limited biocompatibility of metallo-supramolecular assemblies stem from their organic ligand-dominated composition, which restricts interactions with aqueous biological systems. Therefore, we selected the amphiphilic polymer DSPE-PEG2000 to encapsulate **S1/S2**, thereby preparing water-soluble nanoparticles with favorable biocompatibility.^{48,49} NPs with a diameter of around 65 nm were prepared via the nanoprecipitation method. The average diameter obtained from dynamic light scattering (DLS) measurement was 70 nm (Figure 3c,d), in accordance with the transmission electron microscope (TEM) images. It should be noted that due to the swollen effect in aqueous solution, the size obtained from DLS study was always little higher than that obtained from TEM. To evaluate the light-harvesting capability of the nanoparticle formulations, the absorption spectra of **S1@DSPE-PEG2000** and **S2@DSPE-PEG2000** were measured in aqueous solution (Figures S24 and S25). Both nanoparticles exhibit maximum absorption peaks at around 718 nm, which are consistent with their corresponding assemblies in DMSO, confirming that the decker architecture is preserved upon encapsulation. The molar absorption coefficients (ϵ) were calculated to be $1.03 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$ for **S1@DSPE-PEG2000** and $6.09 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$ for **S1@DSPE-PEG2000**. These values confirm their good near-infrared light absorption capacity, supporting their potential for imaging and phototherapy applications. A comparative evaluation of the fluorescence properties and photostability of the encapsulated supramolecular probes was conducted. We systematically assessed the optical performance of DSPE-PEG2000-encapsulated dyes across the >900 nm window, with parallel measurements of the free dyes in DMSO serving as controls (Figure 3e). Encapsulation-induced aggregation led to significant fluorescence quenching relative to the free dyes in DMSO, a trend consistent with the ACQ effect. Notably, despite the superior fluorescence intensity exhibited by **S1** in DMSO, the DSPE-PEG2000-encapsulated nanoparticles formed from **S2**, which possesses a higher TPE content, display markedly enhanced luminescent performance. In stark contrast, **BDP** showed negligible fluorescence under identical laser irradiation conditions, both in DMSO and following DSPE-PEG2000 encapsulation. Quantitative analysis further confirmed that a clear brightness retention hierarchy of **S2** > **S1** > **BDP**: **S2@DSPE-PEG2000** retained approximately 20% of its native brightness (relative to free **S2** in DMSO), whereas **S1** only achieved a 9.3% retention, a direct outcome of the gradient TPE content, where the highest TPE loading in **S2** affords the most efficient ACQ suppression. This finding further confirms that **S1** and **S2** are far more suitable than **BDP** for bioimaging applications. Photostability assessments under a continuous 755 nm laser irradiation revealed striking differences between formulations (Figure 3f). Free dyes in DMSO exhibited relatively rapid photobleaching, whereas the DSPE-PEG2000-encapsulated probes maintained stable emission intensity throughout the 120 min NIR irradiation period. These findings demonstrate that nanoparticle encapsulation leads to a significant enhancement in photostability, likely

through steric protection that reduces molecular collisions and associated quenching pathways. The observed photobleaching resistance confirms that these composite probes offer an extended temporal window for imaging applications. Specifically, the maintained fluorescence signal over prolonged periods (≥ 120 min) under NIR excitation makes these probes particularly suitable for longitudinal NIR-II imaging and monitoring studies where sustained signal stability is paramount.

In biomedical applications, precise visualization of the lymphatic and vascular systems is essential for the functional assessment and regenerative capacity monitoring. The vascular system plays a critical physiological role in nutrient delivery and organ protection, while high-resolution angiography enables surgeons to accurately identify vascular lesions and evaluate postoperative hemodynamic recovery.⁵⁰ To explore these capabilities, we investigated the NIR-II angiographic performance of **dyes@DSPE-PEG2000** in the hindlimb imaging. Following intravenous administration in mice, we systematically evaluated the imaging quality and contrast of **dyes@DSPE-PEG2000** across multiple sub-NIR-II spectral windows (Figure 4a). The imaging quality of the leg vasculature was further assessed by measuring the signal-to-background ratio (SBR), a key metric for evaluating imaging clarity. Quantitative analysis using region-of-interest (ROI) measurements revealed progressive enhancement of vascular contrast when imaging at a longer sub-NIR-II wavelength (900–1300 nm) (Figure 4b). Specifically, the vessel-to-muscle ratio increased from 2.3 (greater than 900 nm) to 3.5 (greater than 1300 nm) for **S1@DSPE-PEG2000**, while **S2@DSPE-PEG2000** demonstrated superior performance with a significantly higher SBR of 5.6 at 1300 nm. The enhanced performance of **S2@DSPE-PEG2000** may be attributed to the lower skin absorption of **S2** and its higher fluorescence retention rate compared with **S1**. The **S2@DSPE-PEG2000** NPs stably emitted sufficiently strong signals in the biological environment, enabling the clear delineation of vascular structures while minimizing background interference from the skin and surrounding tissues. The improved contrast was particularly advantageous for visualizing fine vascular networks in the hind limb model.

Lymphangiography has emerged as an indispensable diagnostic and therapeutic modality for lymphatic system evaluation, providing unique visualization capabilities that complement conventional imaging techniques.⁵¹ This specialized approach enables precise identification of lymphatic disorders, including lymphedema, lymphatic leaks, and chylous effusions, through direct visualization of lymphatic vessels and nodes. With the advancement of personalized medicine, lymphangiography continues to play a pivotal role in managing lymphatic circulation disorders and facilitating patient-specific therapeutic strategies. To systematically evaluate the NIR-II bioimaging performance of **dyes@DSPE-PEG2000** probes, we conducted intradermal injections in the footpads of shaved C57 mice and acquired imaging signals under a >1200 nm window. Remarkably, the **dyes@DSPE-PEG2000** probes maintained exceptional lymphatic imaging performance under the 1200 nm window (Figures 4c and S26), significantly enhancing lymphatic system visualization quality. Quantitative analysis of selected ROIs demonstrated that **dyes@DSPE-PEG2000** consistently exhibited superior imaging capabilities for both lymphatic vessels and lymph nodes (Figure 4d). Notably, the sacral and popliteal lymph nodes were clearly

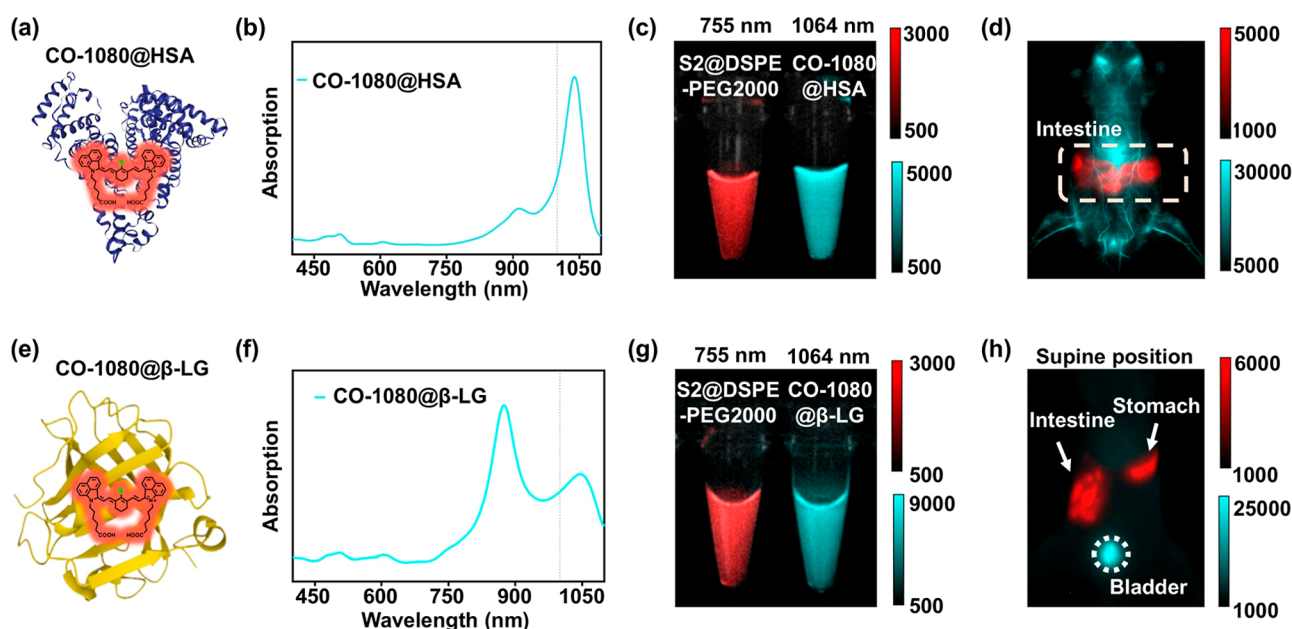


Figure 5. S2@DSPE-PEG2000 and CO-1080@HSA (or CO-1080@ β -LG) for dual-color in vivo imaging. Schematic of (a) the CO-1080@HSA and (e) CO-1080@ β -LG probe. Absorption of the (b) CO-1080@HSA and (f) CO-1080@ β -LG. Brightness of S2@DSPE-PEG2000 (60 ms, 60 μ M)/CO-1080@HSA (60 ms, 20 μ M) and (d) S2@DSPE-PEG2000 (60 ms, 60 μ M)/CO-1080@ β -LG (60 ms, 20 μ M) that can be excited at two separate wavelengths and used for dual-color imaging. (d) Dual-color imaging of intestine and whole-body blood vessels using S2@DSPE-PEG2000 (500 ms, 200 μ M, 200 μ L, gastric lavage) and CO-1080@HSA (500 ms, 600 μ M, 200 μ L, tail vein injection). (h) Dual-color imaging of digestive system and urinary system using S2@DSPE-PEG2000 (>1200 nm, 500 ms, gastric lavage) and CO-1080@ β -LG (>1200 nm, 100 ms, 200 μ M, 200 μ L, tail vein injection). All imaging was performed at the >1200 nm window.

delineated following the probe administration. In addition, hematoxylin and eosin (H&E) staining of major organs (heart, liver, spleen, lungs, and kidneys) confirmed that these nanoparticles exhibit low systematic toxicity to normal organs and tissues (Figure 4e).

Multichannel imaging has become an indispensable tool in modern medical therapy, offering transformative capabilities that enhance diagnostic accuracy, treatment planning, and therapeutic monitoring.^{52,53} By simultaneously capturing multiple spectral or functional signals, this approach provides clinicians with comprehensive, real-time insights that significantly improve patient care. Integrated fluorescence imaging for multiple biological events relies on non-cross-talk fluorescence probes. For dual-color imaging, we selected protein-based CO-1080@HSA (or CO-1080@ β -LG) (Figure 5a,b,e,f) and dyes@DSPE-PEG2000 probes, which exhibited nonoverlapping emissions when excited at 755 and 1064 nm, respectively (Figure 5c,g), demonstrating excellent potential for dual-color imaging.

For multimodal imaging, mice first received an intradermal footpad injection of dyes@DSPE-PEG2000, followed by intravenous administration of CO-1080@HSA. Signals were acquired sequentially under 755 and 1064 nm laser excitation (Figure S27a,c,d). This approach revealed high-resolution vascular distribution around lymphatic vessels and nodes, which was critical for image-guided surgical precision. Following this, we utilized the selected fluorescent probes to accomplish simultaneous dual-channel imaging of the whole-body blood vessel network and intestinal tract (Figure 5d). We further employed S2@DSPE-PEG2000 and CO-1080@ β -LG to achieve crosstalk-free bioimaging of both the digestive and urinary systems (Figures 5h and S27b).

CONCLUSION

In summary, this work presents a decker supramolecular assembly strategy to mitigate the critical ACQ issue that limits the application of aza-BODIPY fluorophores in NIR bioimaging. By tailoring the stoichiometric ratio of sterically hindered TPE units to ACQ-sensitive aza-BODIPY moieties, we constructed three distinct systems with gradient TPE contents: BDP (0% TPE), triple-decker S1 (33% TPE), and double-decker S2 (50% TPE). The propeller-shaped TPE moieties spatially occupy one face of aza-BODIPY, precisely modulating intermolecular π - π stacking and enabling graded ACQ suppression with an efficacy order of S2 > S1 > BDP. Systematic photophysical characterization validated the superior performance of the decker architectures: S2@DSPE-PEG2000 NPs retained ~20% of their native brightness in the aggregated state (vs 9.3% for S1 and negligible retention for BDP), achieved red-shifted absorption, and exhibited excellent photostability under continuous NIR irradiation. In biologically relevant spectral windows, these NPs enabled high-contrast, real-time visualization of deep-tissue vasculature and lymphatic networks, demonstrating their enhanced applicability for in vivo bioimaging compared to conventional aza-BODIPY probes. This study not only elucidates the structure-property relationship between fluorophore stacking and ACQ suppression but also provides a novel paradigm for overcoming aggregation-caused limitations in organic fluorophores for bioimaging.

MATERIALS AND METHODS

General Procedures

All reagents were purchased from Macklin Reagent, Energy Chemical, and Alfa Aesar and used without further purification. 4'-Bromo-2,2':6',2''-terpyridine was purchased from the Jilin Chinese Academy

of Sciences—Yanshen Technology Co. Ltd. Column chromatography was conducted using SiO₂ (VWR, 40–60 μ m, 60 Å), and the separated products were visualized by UV light.

Nuclear Magnetic Resonance

NMR spectral data were recorded on a Qone AS 400 MHz, Bruker 500 MHz, and 600 MHz Advance NMR spectrometer in CDCl₃ or CD₃CN with TMS as reference.

Electrospray Ionization-Mass Spectrometry and Traveling Wave Ion Mobility-Mass Spectrometry

Electrospray ionization (ESI) mass spectra were recorded with a Waters Synapt G2 tandem mass spectrometer, using solutions of 0.5 mg sample in 1 mL of MeCN/MeOH (3:1, v/v) for complexes. The TWIM-MS experiments were performed under the following conditions: ESI capillary voltage, 3 kV; sample cone voltage, 30 V; extraction cone voltage, 3.5 V; source temperature 100 °C; desolvation temperature, 100 °C; cone gas flow, 10 L/h; desolvation gas flow, 700 L/h (N₂); source gas control, 0 mL/min; trap gas control, 2 mL/min; helium cell gas control, 100 mL/min; ion mobility (IM) cell gas control, 30 mL/min; sample flow rate, 5 μ L/min; IM traveling wave height, 25 V; and IM traveling wave velocity, 1000 m/s.

Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry

MALDI-TOF-MS was performed on a Bruker AutoflexIII using *trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyli-dene]malononitrile (DCTB) as a matrix. The matrix was dissolved in CHCl₃ at 20 mg mL⁻¹, and organic compounds were dissolved in CHCl₃ at 10 mg mL⁻¹. The sample was prepared by depositing 0.5 μ L of matrix on the wells of a 384-well ground-steel plate, allowing the spots to dry, depositing 0.5 μ L of the sample on a spot of dry matrix, and adding another 0.5 μ L of matrix on top of the dry sample. The plate was inserted into the MALDI source after drying. The sample was conducted in linear mode. And the data analysis was conducted with Bruker's FlexAnalysis software.

Computational Details

All calculations were carried out with the Gaussian 16 software.⁹ Density functional theory (DFT) calculations were carried out with the PBE0 functional with the combination of the Grimme's D3BJ version of dispersion correction. The basis set of 6-31G (d) was adopted for the geometry optimization and frequency calculations. The geometries were fully optimized without any structural constraints. The implicit solvation model SMD with acetonitrile as the solvent was applied in the calculations. The harmonic frequency calculations were carried out at the same level of theory to verify that all structures have no imaginary frequency.

Photophysical Measurement

UV–vis spectra of solutions were recorded on a PerkinElmer LAMBDA 365 spectrophotometer. Fluorescence emission spectra were measured by a Shimadzu spectrofluorimeter RF-5301PC. Absolute PL quantum yields and fluorescence lifetimes were carried out with an FLS 980 Spectrometer (Edinburgh Instruments Ltd., Livingston, United Kingdom). The absolute PL quantum yield was measured via an integrating light sphere. Solutions were placed in 1 cm path length quartz cells. The absorption and emission spectra of different probes were measured using a PerkinElmer Lambda 950, an Edinburgh Instruments FL 920, and a MARS-IRIS 400 to a 1700 nm absorbance/fluorescence spectrophotometer.

Preparation of Supramolecular Assembly-Loaded Nanoparticles

DSPE-PEG2000 was first dissolved in DMSO to form a stocker (100 mM). 25 μ L of the stocker and 2.5 μ L of dye (S1 or S2; 20 mM in DMSO) were mixed by the vortex. The mixture was added to PBS to form a 10 mL (5 μ M) solution. The aqueous solution was further sonicated in an ice–water bath for 30 min and then incubated in a water bath at 50 °C for 1 h. After being stored at room temperature

for 1 h in the dark, the solution was concentrated by ultrafiltration for NIR-II bioimaging.

Transmission Electron Microscope

The sample was dissolved in PBS at a concentration of 20 μ M. The solutions were drop-cast on to a carbon-coated Cu grid, and extra solution was absorbed by filter paper to avoid aggregation. The TEM images of the drop-cast samples were taken with a JEOL 2010 transmission electron microscope.

Animal Experiments

All animal experiments were conducted under institutional guidelines and were approved by the Animal Ethical Committee of The First Hospital of Jilin University (20210642). Balb/c and C57 mice were purchased from Liaoning Changsheng Biotechnology Co. Ltd. and Beijing Vital River Laboratory Animal Technology Co., Ltd., respectively. Bedding, nesting materials, food, and water were provided ad libitum. The ambient temperature was controlled at 20–24 °C with 12 h light/12 h dark cycles.

NIR-II Fluorescence Imaging

All mice were shaved using Nair hair removal cream (Nair Lotion with Aloe & Lanolin) and anesthetized with isoflurane before the experiment. All images were collected on an InGaAs camera (Princeton Instruments, NIRvana-640) with a laser of 755 or 1064 nm. At a 755 nm excitation, the power density for NIR-II brightness measurement and NIR-II whole-body bioimaging was set to 38 mW/cm², and the power density for NIR-II magnified bioimaging was set to 100 mW/cm². At a 1064 nm excitation, the power density for NIR-II imaging was set to 65 mW/cm². The emission fluorescence was typically collected using different long-pass filters. A variable exposure time was used for the InGaAs camera to capture images in the sub-NIR-II windows. Meanwhile, the images used all C57 mice.

Statistical Analyses

All fluorescence intensity data are expressed as mean $\pm$ SD. For normally distributed data sets with equal variances, two-way ANOVA testing followed by a Tukey post hoc test was carried out across groups. In all cases, significance was defined as ns < 0.05, ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.6c04422>.

Synthesis and structural characterization of ligands and assemblies (including NMR, MALDI-TOF, and ESI-MS spectral data); photophysical properties of ligands and complexes; and photophysical properties and in vivo imaging of nanoprobes (PDF)

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Notes

The authors declare no competing financial interest.

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